

FORGE

Framework for Orchestrated Requirements, Governance & Engineering

A structured software development system for OpenCode

What is FORGE?

FORGE is a comprehensive methodology for AI-assisted software development that combines:

- **Structured progressive context** that prevents inconsistent decisions
- **Adaptive process** that calibrates ceremony to complexity
- **Persistent knowledge** that survives session boundaries

Built natively for OpenCode, synthesising the best of BMAD Method and Speckit

The Problem

5 Critical Problems in AI-Assisted Development

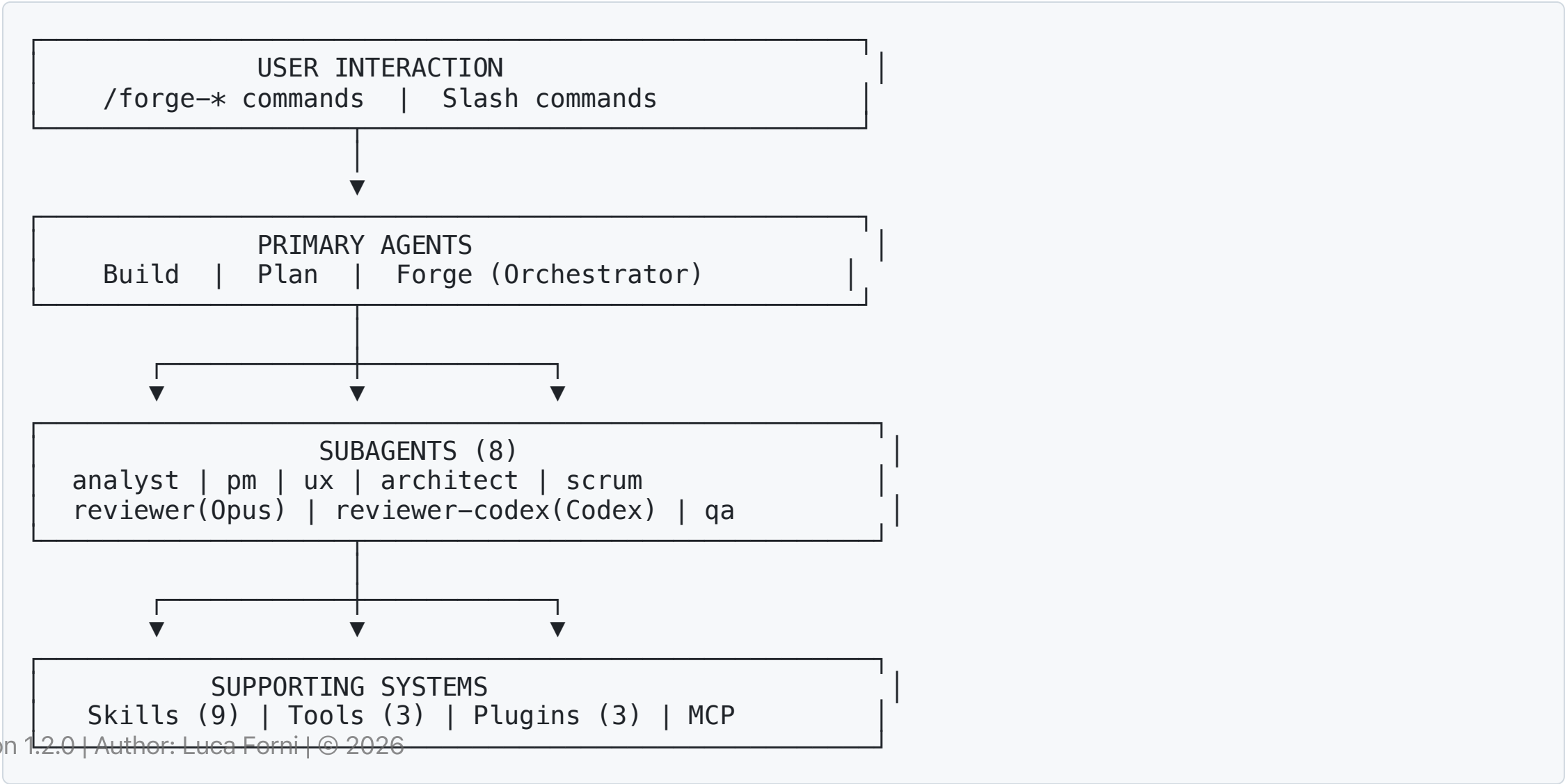
1. **Context Gap** - Agents in different sessions make conflicting decisions
2. **Ceremony Trap** - Too much process kills speed, too little kills quality
3. **Knowledge Evaporation** - Decisions disappear at the end of the session
4. **Consistency Entropy** - Teams of 15+ developers produce inconsistent code
5. **Illusion of Productivity** - Speed without direction is waste

The FORGE Solution

6 Core Principles

Principle	Description
Progressive Context Engineering	Each phase produces a document that becomes context for the next phase
Constitutional Governance	Immutable principles govern every decision
Adaptive Ceremony	Process depth scales with complexity
Adversarial Quality	Reviews MUST find problems
Persistent Knowledge	Decisions and lessons survive across sessions
Bidirectional Traceability	Requirements → code → tests (and back)

System Architecture



5 Workflow Tracks

Complexity —————> High				
Hotfix	Quick	Feature	Epic	Product
1 file	1-5 tasks	5-20 tasks	20-50+	New product
< 30 min	< 1 day	1-5 days	1-4 weeks	4+ weeks
No docs	Tech spec	Spec+Plan	Full chain + Sprint	Full chain + Constitution

The process automatically adapts to task complexity

Track: Hotfix

When: Critical bug, 1-2 files, < 30 minutes

```
/forge-hotfix "Login endpoint returns 500 when user has no profile picture"
```

Workflow:

1. ✓ Diagnose - Identify root cause
2. ✓ Fix - Apply minimal, targeted fix
3. ✓ Verify - Run existing tests
4. ✓ Review - Quick self-review vs constitution

Output: Structured commit message (no additional documents)

Track: Quick

When: Small feature, 1-5 tasks, < 1 day

```
/forge-quick "Add forgot password feature with 1-hour expiry token"
```

Workflow:

1. ✓ Quick Spec - Conversation → `tech-spec.md`
2. ✓ Implement - Implement task by task
3. ✓ Test - Generate unit tests
4. ✓ Review - Adversarial self-review

Output: `.forge/specs/NNN-name/tech-spec.md` + code + tests

Track: Feature

When: Medium feature, 5-20 tasks, 1-5 days

```
/forge-specify "Add OAuth2 authentication with Google and GitHub"  
/forge-clarify    # Resolve ambiguities  
/forge-ux         # UX design: personas, wireframes, accessibility  
/forge-plan       # Technical plan  
/forge-analyze    # Cross-validate spec vs plan  
/forge-tasks      # Task breakdown with dependencies  
/forge-implement  # Implement  
/forge-review     # Adversarial review (7 dimensions, dual-model: Claude Opus + GPT-Codex)
```

Output: spec.md, design-spec.md, user-journey.md, plan.md, tasks.md, optional ADRs

UX Phase: /forge-ux

When: Feature with user interface (web, mobile, design system)

/forge-ux "Login page with OAuth"

Artifact	Content	Focus
design-spec.md	ASCII wireframes, UI components, interaction specs	 wireframes ·  design system ·  responsive
user-journey.md	Personas, scenarios, user flows	 journeys ·  WCAG 2.1 AA

Review: 7 Quality Dimensions

The review covers all 7 dimensions:

Correctness	→ logic, edge cases
Security	→ vulnerabilities, injection
Performance	→ slow queries, memory leaks
Maintainability	→ readability, coupling
Constitution	→ adherence to principles
Test-Spec Coherence	→ tests cover spec requirements
UX Quality	→ accessibility, usability, design consistency

UX issue examples:

- [HIGH] Button without `aria-label` — not accessible with screen reader
- [MEDIUM] Form without focus management — degraded UX after submit
- [LOW] Colour contrast 2.8:1 — below WCAG AA threshold (4.5:1)

Track: Epic

When: Complex feature set, 20-50+ tasks, 1-4 weeks

/forge-brief	# Product brief
/forge-prd	# Full PRD
/forge-architecture	# Architecture + ADR
/forge-sprint	# Sprint planning
/forge-story	# Prepare story
/forge-implement	# Implement story
/forge-review	# Dual review (AI + Human)
/forge-retro	# Retrospective

Output: Brief, PRD, Architecture, Epic, Stories, Sprint Status, ADR

Track: Product

When: New product, greenfield, 4+ weeks

```
/forge-init          # Setup + Constitution
```

Then follows the Epic workflow with the addition of:

- **Constitution** - Immutable governance document
- **UX Phase** - `/forge-ux` for products with user interface

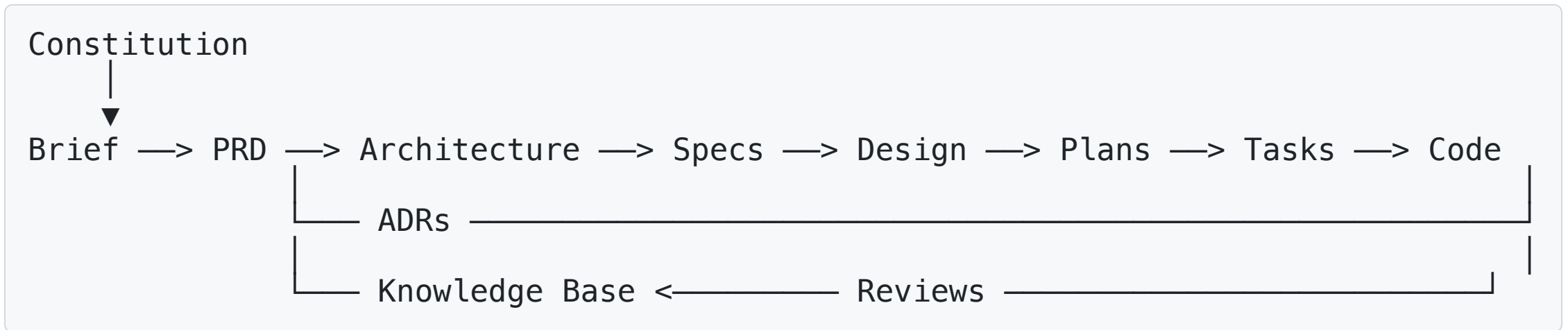
Full output: All Epic track documents + Constitution

The 8 Specialized Subagents

Agent	Model	Role
forge-analyst	Sonnet 4.5	Exploration, research, scope detection
forge-pm	Opus 4.6	Requirements, spec, PRD, user stories
forge-ux	Opus 4.6	User journeys, wireframes, accessibility, design spec
forge-architect	Opus 4.6	Architecture, ADR, technical planning
forge-scrum	Sonnet 4.5	Sprint planning, story management
forge-reviewer	Opus 4.6	Adversarial review — Task A (7 dimensions)
forge-reviewer-codex	GPT-Codex	Adversarial review — Task B, independent
forge-qa	Sonnet 4.5	Test strategy, test generation

`/forge-review` launches **forge-reviewer + forge-reviewer-codex in parallel** and synthesises findings — *consensus findings* (both models) carry the highest priority

Progressive Context Chain



Each phase receives structured context from the previous phase

- Prevents inconsistent decisions
- Guarantees architectural alignment
- Accumulates knowledge over time

Constitutional Governance

The **Constitution** is the highest-authority governance document

9 Standard Articles

1. **Core Principles** - Non-negotiable principles
2. **Technology Stack** - Approved tech stack
3. **Architecture Patterns** - Mandatory patterns
4. **Quality Standards** - Quality thresholds
5. **Security** - Security requirements
6. **Error Handling** - Error handling standards
7. **Naming & Conventions** - Naming conventions
8. **Testing Standards** - Required tests
9. **Operational Requirements** - Operational standards

Persistent Knowledge Base

3 Types of Artefacts

```
.forge/knowledge/  
├── adr/                                # Architecture Decision Records  
│   ├── 001-database-choice.md  
│   └── 002-auth-strategy.md  
├── decision-log.md                    # Session-extracted decisions  
└── lessons-learned.md                 # Post-mortem insights
```

Features:

- ✓ Created automatically by the `session-knowledge` plugin
- ✓ Persist across sessions
- ✓ Loaded as context every session
- ✓ Prevent repeated mistakes

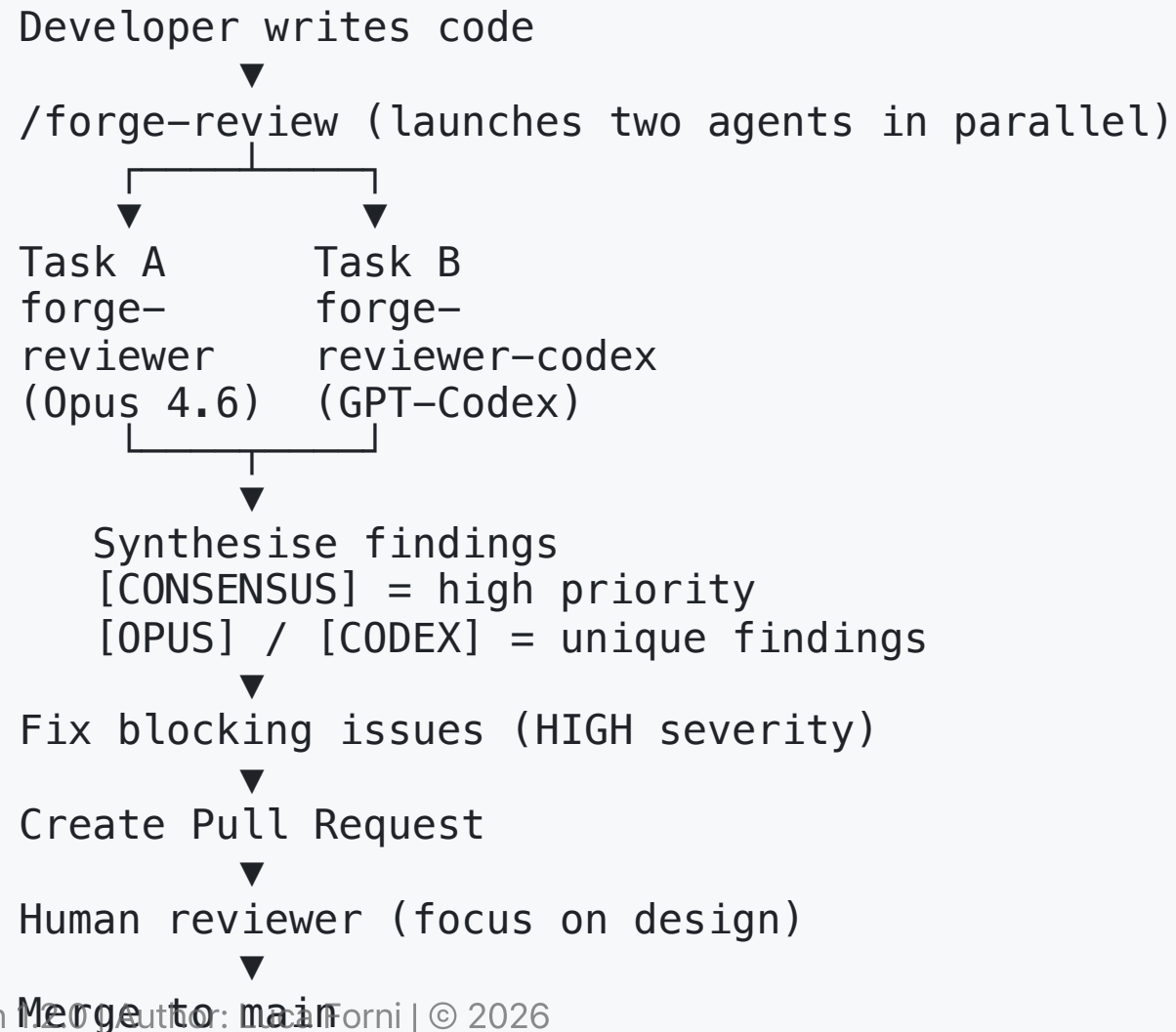
Adversarial Review

Two Models in Parallel — Reviews that **MUST** Find Problems

`forge-reviewer` (Claude Opus) + `forge-reviewer-codex` (GPT-Codex) examine across **7 dimensions**:

1. **Correctness** - Logic, edge cases, errors
2. **Security** - Vulnerabilities, input validation
3. **Performance** - Slow queries, memory leaks
4. **Maintainability** - Readability, coupling
5. **Constitution Compliance** - Adherence to principles
6. **Test-Spec Coherence** - Tests cover spec requirements
7. **UX Quality** - Accessibility (WCAG 2.1 AA), usability, design consistency

Dual Review Process



Bidirectional Traceability

```
Requirement FR-001 (spec.md)
↓
Technical approach (plan.md, section 3.2)
↓
Task 2.1 (tasks.md)
↓
Source file (src/auth/login.ts)
↓
Test file (src/auth/__tests__/login.test.ts)
```

Custom tool: `/forge-analyze` generates traceability matrix

- Identifies unimplemented requirements
- Identifies code without requirements (orphan code)
- Verifies test coverage per requirement

Brownfield Support

Onboarding Existing Codebases

```
/forge-init # In existing project
```

FORGE analyses:

- Project structure and languages
- Detected frameworks and architectural patterns
- Naming and organisation conventions
- Potential tech debt

Output:

- Constitution auto-generated based on existing code
- **AGENTS.md** derived from `.eslintrc`, `tsconfig.json`, detected patterns

9 Dynamic Skills

Skill	Used by	Purpose
adversarial-review	forge-reviewer	Mandatory review protocol
advanced-elicitation	pm, architect	Deep analysis techniques
scope-detection	Forge orchestrator	Assess complexity, recommend track
test-strategy	forge-qa, Build	Adaptive test strategy per track
brownfield-analysis	forge-analyst	Existing codebase analysis
constitution-compliance	architect, reviewer	Verify constitution compliance
context-chain	All agents	Load correct upstream documents
ux-design	forge-ux	Generate user journeys, wireframes, a11y specs
ux-review	forge-reviewer	7th review dimension: UX quality & accessibility

Loaded on-demand to save context window

Automatic Plugins (1/2)

session-knowledge

- Automatically extracts decisions and lessons
- Appends to `decision-log.md` and `lessons-learned.md`
- Injects knowledge during context compaction

pre-commit-gate

- Validates spec-code consistency before commit
- Verifies completed tasks and existing tests
- Advisory (does not block, but warns)

Automatic Plugins (2/2)

spec-watcher

- Monitors changes to `.forge/specs/`
- Detects inconsistencies with plan/tasks
- Suggests `/forge-analyze`

Plugins automate governance and knowledge management without manual intervention

Benefits for Developers

Benefit	How FORGE Ensures It
Fast onboarding	Spec, plan, ADR provide full context
Quality on the first try	Structured context → better AI decisions
Less rework after review	AI review catches issues before human review
Clear requirements	/forge-clarify surfaces ambiguities before implementation
Confidence in decisions	Constitution and ADR validate choices

Benefits for Teams

Benefit	How FORGE Ensures It
Consistent code	Constitution enforces patterns; all agents follow the same rules
Reduces architecture drift	ADRs prevent contradictory decisions
Efficient sprint management	<code>/forge-sprint</code> and <code>/forge-status</code> automate overhead
Effective retrospectives	<code>/forge-retro</code> produces actionable insights
Faster reviews	AI handles mechanical checks, humans focus on design
Smooth onboarding	New members read <code>.forge/</code> to understand the whole project
Safe parallel development	Architecture defines boundaries; ADRs prevent conflicts

Benefits for Enterprise

Benefit	How FORGE Ensures It
Audit trail compliance	Constitution + ADR + decision log = complete rationale
Risk management	Spec includes risk sections; adversarial review finds issues early
Knowledge retention	When developers leave, decisions remain
Process standardisation	All teams use the same workflows, templates, protocols
Tech debt visibility	Brownfield analysis + traceability expose gaps
Quality metrics	Sprint velocity, issue count, spec completeness
Governance without bottleneck	Constitution + skills enforce standards automatically

FORGE vs Alternatives

vs No Methodology ("Vibe Coding")

	No Methodology	FORGE
Initial speed	Very fast	Moderate (planning overhead)
Speed over time	Slows down (debt accumulates)	Sustained (compounding knowledge)
Consistency	Random	Enforced
Rework	High (30-50%)	Low (< 15%)
Onboarding	Weeks	Days

FORGE vs BMAD vs Speckit

	BMAD	Speckit	FORGE
Agent system	Personas (1 LLM)	None	Real subagents
Tracks	3	1	5
Governance	Weak	Constitution	Constitution + ADR + KB
Knowledge persistence	None	Per-branch	Cross-session KB
Review	Adversarial	None	Dual Adversarial
Brownfield	Limited	Limited	Structured
Platform	IDE-agnostic	Agent-agnostic	OpenCode-native

When to Use FORGE

 Ideal Use Cases

Scenario	Track	Why
New SaaS from scratch	Product	Prevents tech debt
Enterprise feature	Epic	Prevents scope creep
Large codebase	Feature/Quick	Maintains consistency
Compliance-heavy	Epic/Product	Complete audit trail
Team with turnover	Any	Retains knowledge

When NOT to Use FORGE

✗ Overkill Scenarios

Scenario	Better Alternative
One-off scripts or utilities	Write code directly
Tutorial / learning projects	Focus on learning, not process
Hackathon prototype (discard after)	Speed > structure
Projects with < 1 week total lifespan	The docs outlive the code
Safety-critical systems	FORGE + formal verification
Multi-team platform (50+ devs)	FORGE + enterprise program management

The Cost of Structure

Realistic Overhead

Activity	Overhead
Constitution	1-2 hours (once)
Spec (Feature)	20-30 min
Plan	15-20 min
Analyze	5 min
Review	10-15 min
Retro	15-20 min/sprint

Total per feature: ~1-1.5 hours

Break-Even Analysis

Without FORGE:

Implementation: 3 days

Rework after review: 0.5 days (frequent)

Rework after production: 1 day (occasional)

Total: 3.8 days

With FORGE:

Planning + review: 0.2 days

Implementation: 2.5 days (better context = faster)

Rework after review: 0.1 days (AI review catches issues)

Rework after production: 0.1 days (rare with dual review)

Total: 2.9 days

Break-even typically at the 2nd–3rd feature of the project

Compound Effect

Session 1: You pay the constitution cost → high overhead, zero benefit
Session 10: Constitution prevents a bad decision → benefit > costs
Session 50: New dev productive in hours thanks to .forge/
Session 200: Auditor asks encryption rationale → ADR-012 → 1 day vs 1 week

Structure has decreasing costs and compounding returns

The question is not "can we afford the overhead?" but
"can we afford NOT to have it?"

Technical Components

10 Agents | 21 Commands | 9 Skills

```
.opencode/  
├── agents/           # 8 specialized subagents + forge orchestrator + Build + Plan  
├── commands/        # 21 slash commands  
├── skills/           # 9 dynamic skills  
├── tools/            # 3 custom tools  
├── plugins/          # 3 automation plugins  
└── templates/        # 11 document templates
```

All configured in `opencode.json`

Quick Start

1. Installation

```
# Copy .opencode/ into your project
cp -r path/to/forge/.opencode/ your-project/.opencode/
cp path/to/forge/opencode.json your-project/opencode.json
```

2. Verify

```
cd your-project
opencode
> /forge-help
```

3. First Workflow (Quick)

```
> /forge-quick "Add health check endpoint returning app version"
```

Directory Structure



Model Strategy

Differentiation by Cognitive Demand

Model	When	Agents
Claude Opus 4.6	Deep reasoning, architectural decisions, adversarial review	forge-pm, forge-architect, forge-reviewer
Claude Sonnet 4.5	Speed, good-enough reasoning, analysis, sprint mgmt	Forge, forge-analyst, forge-scrum, forge-qa, Build, Plan
GPT-5.2-Codex	Independent adversarial review (second model)	forge-reviewer-codex

Models provided via GitHub Copilot subscription

Incremental Adoption Path

No Need to Adopt Everything at Once

Week 1-2: Hotfix + Quick only

- Get familiar with the workflow without changing existing process

Week 3-4: Feature track for a medium feature

- Experience the full spec-plan-implement-review cycle

Month 2: Epic track for a larger initiative

- Add sprint management

Month 3+: Evaluate Product track and constitutional governance

Essential Commands

Quick Reference Card

Tracks:	<code>/forge-hotfix</code>	Bug fix, 1 file, < 30 min
	<code>/forge-quick</code>	Small feature, 1-5 tasks
	<code>/forge-specify</code>	Medium feature (start here)
	<code>/forge-brief</code>	Large epic
	<code>/forge-init</code>	New product
Feature Flow:	<code>specify → clarify → ux → plan → analyze</code> <code>→ tasks → implement → review</code>	
UX Commands:	<code>/forge-ux</code>	User journeys, wireframes, a11y
	<code>/forge-wireframe</code>	ASCII wireframe generation
Status:	<code>/forge-status</code>	Sprint dashboard
	<code>/forge-help</code>	Context-aware help
Knowledge:	<code>/forge-adr</code>	Create ADR
	<code>/forge-retro</code>	Sprint retrospective

Use Case: OAuth Feature

```
# 1. Specify
/forge-specify "Add OAuth2 auth with Google and GitHub"

# 2. Clarify ambiguities
/forge-clarify

# 3. UX Design
/forge-ux # login wireframe, user journey, a11y

# 4. Technical plan
/forge-plan

# 5. Validate consistency
/forge-analyze

# 6. Task breakdown
/forge-tasks

# 7. Implement
/forge-implement

# 8. Adversarial review (7 dimensions, dual-model: Claude Opus + GPT-Codex)
/forge-review
```

Team Workflow

Multi-Developer Feature Development

Developer A: Epic 1, Stories S001–S004 (Core Payments)
Developer B: Epic 2, Stories S001–S003 (Subscriptions)
Developer C: Epic 3, Stories S001–S003 (Webhooks)

Shared artefacts (via git):

- `.forge/constitution.md` - Everyone follows the same principles
- `.forge/architecture/` - Consistent technical decisions
- `.forge/knowledge/adr/` - Everyone sees all decisions
- `.forge/sprints/` - Scrum master updates centrally

Conflict prevention:

Sprint Ceremonies with FORGE (1/2)

Sprint Planning

```
/forge-sprint  
# → Review velocity  
# → Select stories from backlog  
# → Assign to developer  
# → Update sprint-status.yaml
```

Daily Standup

```
/forge-status  
# → See assigned stories and status  
# → Flag blockers
```

Sprint Ceremonies with FORGE (2/2)

Sprint Retrospective

```
/forge-retro  
# → Automatic velocity calculation  
# → Lessons learned → knowledge base
```

Knowledge Base Maintenance

Task	Frequency	Owner
Review decision-log, promote to ADR	Weekly	Tech Lead
Archive stale lessons-learned	Per sprint	Scrum Master
Review ADR staleness	Monthly	Architect
Verify constitution accuracy	Quarterly	Tech Lead + PM

15 minutes/week for KB maintenance

Customisation

FORGE is Fully Customisable

- **Constitution Template** - Adapt articles to your domain
- **Agent System Prompts** - Tune for team style
- **Skills** - Add domain-specific reasoning
- **Templates** - Modify document structure
- **Commands** - Create custom workflows
- **Plugins** - Project-specific automation

See: `.opencode/docs/FORGE-CUSTOMIZATION.md`

CI/CD Integration

Pre-commit Gate Plugin

```
// Runs before commit
- Check tasks completed in tasks.md
- Verify tests exist for modified files
- Validate [NEEDS CLARIFICATION] resolved
- Check constitution compliance verified
```

Non-blocking (advisory), but provides a clear signal

Advanced Elicitation Techniques

6 Deep Analysis Techniques

1. **Pre-mortem Analysis** - Imagine the feature failed — what went wrong?
2. **First Principles Thinking** - Decompose into fundamental components
3. **Red Team / Blue Team** - Attacker vs defender
4. **Socratic Questioning** - Deep questions about assumptions
5. **Constraint Removal** - What if we had no constraints?
6. **Inversion Analysis** - How would we guarantee it FAILS?

Used by `forge-pm` and `forge-architect` for spec and architecture analysis

Scope Detection

Automatic Complexity Assessment

7 Factors:

- Files affected (1-2 → 50+)
- Estimated tasks (1 → 50+)
- New dependencies (0 → Stack decision)
- Schema changes (None → Full design)
- API surface changes
- Cross-module impact
- Need for new patterns

Output: Structured JSON with recommended track + reasoning

Adaptive Test Strategy

Coverage Requirements per Track

Track	Required Tests
Hotfix	Regression test for the bug only
Quick	Unit tests for new/modified code
Feature	Unit + Integration tests
Epic	Unit + Integration + E2E
Product	Unit + Integration + E2E + Performance benchmarks

Defined by the `test-strategy` skill, executed by `forge-qa`

ADR (Architecture Decision Records)

When to Create an ADR

- Database, framework, major library choice
- API style decision (REST vs GraphQL vs gRPC)
- Defining pattern (event-driven vs synchronous)
- Trade-off (consistency vs availability)
- Any decision someone will question later

Format

```
# Context: Why is this decision needed?  
# Options: What alternatives were considered?  
# Decision: What was chosen and why?  
# Consequences: Positive, negative, neutral effects  
# Constitution Alignment: Which articles does it support?
```

Retrospectives

/forge-retro Output

Sprint 2 Retrospective

=====

Velocity: 32 pts (planned: 34, previous: 28)

What went well:

- Stripe integration straightforward thanks to clear ADRs
- Task parallelism markers saved time

What could improve:

- Webhook testing required manual Stripe CLI setup (not in spec)
- OAuth PR took 2 days human review (bottleneck)

Action items:

- Add webhook test setup instructions to spec template
- Rotate PR reviewer to avoid single-person bottleneck

Lessons → `forge/knowledge/lessons-learned.md`

Traceability Matrix Example

```
Requirement FR-001 (Login with OAuth)
  → Plan Section 3.2 (OAuth flow implementation)
    → Task 2.1 (Implement OAuth strategy)
      → src/auth/oauth-strategy.ts ✓
      → src/auth/__tests__/oauth-strategy.test.ts ✓

Requirement FR-002 (Account linking)
  → Plan Section 3.3 (Link accounts)
    → Task 3.1 (Link endpoint) → [NOT IMPLEMENTED] ⚠

Requirement NFR-001 (P95 < 200ms)
  → [NO TASK] → [NOT IMPLEMENTED] ⚠
```

Generated by the **trace-requirements** tool

Success Metrics

How to Measure FORGE Success

Metric	Target	How Measured
Rework rate	< 15%	Issues after PR merge / Total PRs
Onboarding time	< 3 days	New dev to first meaningful commit
Architecture drift incidents	0	ADR violations detected
Knowledge retention	100%	Documented decisions / Total decisions
Sprint velocity	+20% after 3 sprints	Story points delivered
Code consistency	> 90%	Constitution compliance score

Resources & Documentation

Full Documentation in `.opencode/docs/`

- [FORGE-GUIDE.md](#) - Complete usage guide
- [FORGE-PHILOSOPHY.md](#) - Principles and benefits
- [FORGE-PROJECT-PLAN.md](#) - System architecture
- [FORGE-CUSTOMIZATION.md](#) - How to customise
- [FORGE-DECISIONS.md](#) - Methodology decision records

Community & Support

- GitHub: `lucaforni/forge`
- OpenCode Docs: `https://opencode.ai/docs`

Future Roadmap

Possible Evolutions

- **AI Pair Programming Mode** - Forge assists in real-time during coding
- **Multi-repo Support** - FORGE coordination across microservices
- **Custom Domain Templates** - Pre-built constitutions for fintech, healthcare, gaming
- **Integration with Project Management Tools** - Jira, Linear, Azure DevOps
- **Visual Architecture Diagrams** - Auto-generate from architecture.md
- **Metrics Dashboard** - Real-time visibility on velocity, quality, debt

Key Takeaways

Why FORGE Changes the Game

1. **Structured context** prevents inconsistencies across AI sessions
2. **5 workflow tracks** adapt the process to complexity
3. **UX-first design** includes user journeys, wireframes and accessibility in the workflow
4. **Constitutional governance** ensures a uniform quality bar
5. **Persistent knowledge base** eliminates knowledge loss
6. **Adversarial review** catches issues before production (7 dimensions, dual-model)
7. **Native OpenCode integration** leverages the full platform
8. **Brownfield support** onboards existing codebases
9. **Team-ready** supports 15+ developers with parallel development

Get Started Today (1/2)

3 Steps to Try FORGE

1. Setup (5 minutes)

```
cp -r .opencode/ your-project/  
cd your-project && opencode
```

2. First Task (5 minutes)

```
> /forge-quick "Your first small feature"
```

Get Started Today (2/2)

3 Steps to Try FORGE

3. Evaluate Results

- Is the produced spec clear?
- Is the implementation clean?
- Were tests generated correctly?

If yes → Continue with Feature track

If no → Contact support for tuning

Questions?

Contacts:

- Documentation: `.opencode/docs/`
- GitHub Issues: `lucaforni/forge`
- OpenCode Docs: `https://opencode.ai/docs`

Thank You!

FORGE

Framework for Orchestrated Requirements, Governance & Engineering

Enterprise software development with AI — structured and sustainable

Author: Luca Forni

 [linkedin.com/in/lucaforni](https://www.linkedin.com/in/lucaforni)

 github.com/lucaforni

Version 1.2.0 / MIT License / 2026